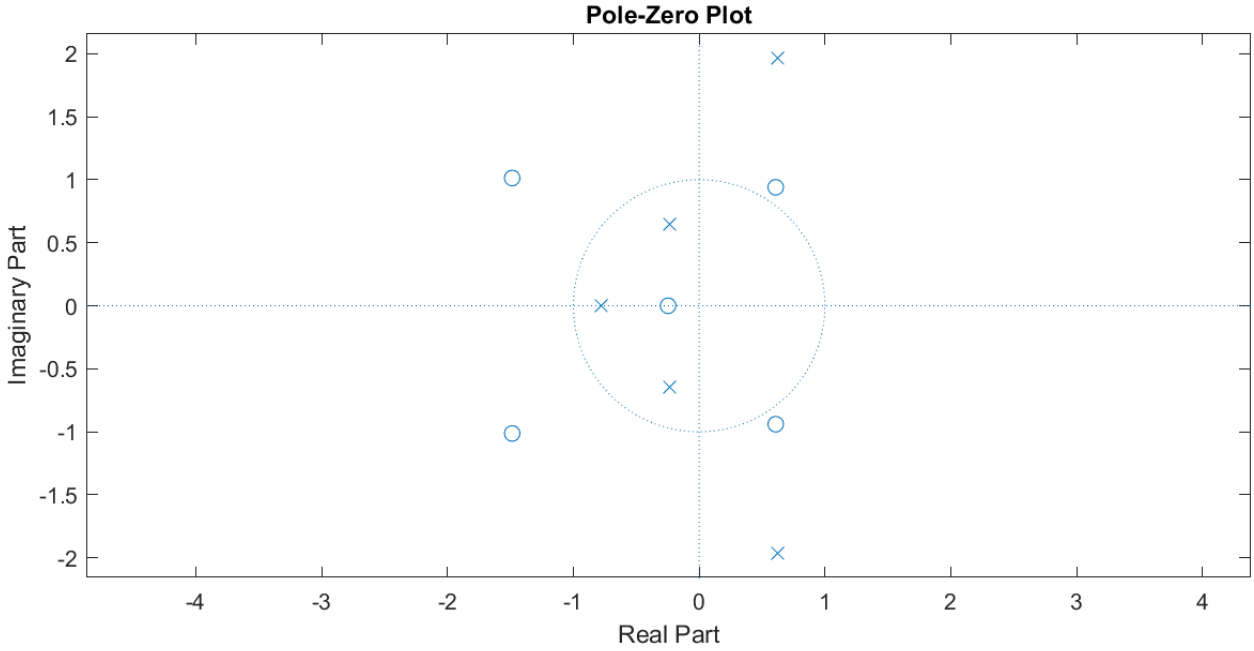
**Code:**

1. Z – Transform:

```
% z-transform
num=[1,2,1.3,0,4,1];
den=[0.65,0,2.3,3,2,1];
[z,p,k]=tf2zp(num,den);
zplane(z,p)
```




```
%z-transform
z =[
  -1.4857 + 1.0130i
  -1.4857 - 1.0130i
   0.6090 + 0.9394i
   0.6090 - 0.9394i
  -0.2467 + 0.0000i
]; % zeros
p =[
   0.6212 + 1.9665i
   0.6212 - 1.9665i
  -0.7748 + 0.0000i
  -0.2338 + 0.6420i
  -0.2338 - 0.6420i
]; % poles
k =1.5385; % gain
[num,den]=zp2tf(z,p,k);
% Normalize
num = num / den(1);
den = den / den(1);
% Display H(z)
Hz = tf(num, den, -1)
```

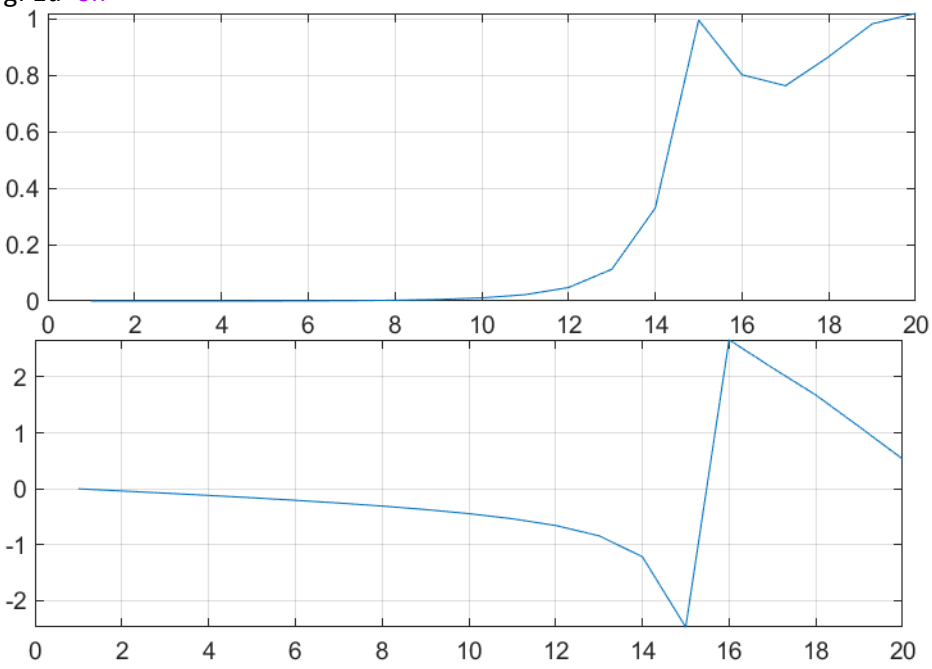
Output:

Hz =

$$\frac{1.538\,z^5 + 3.077\,z^4 + 2\,z^3 - 0.0001606\,z^2 + 6.154\,z + 1.538}{z^5 + 1.11e\text{-}16\,z^4 + 3.539\,z^3 + 4.616\,z^2 + 3.077\,z + 1.538}$$

2. Frequency Response:

```
% frequency response
num=[0.008,-0.033,0.05,-0.033,0.008]
den=[1,2.37,2.7,1.6,0.4]
w=20 % range of frequency
x=freqz(num,den,w)
subplot(211)
plot(abs(x))
grid on
subplot(212)
plot(angle(x))
grid on
```



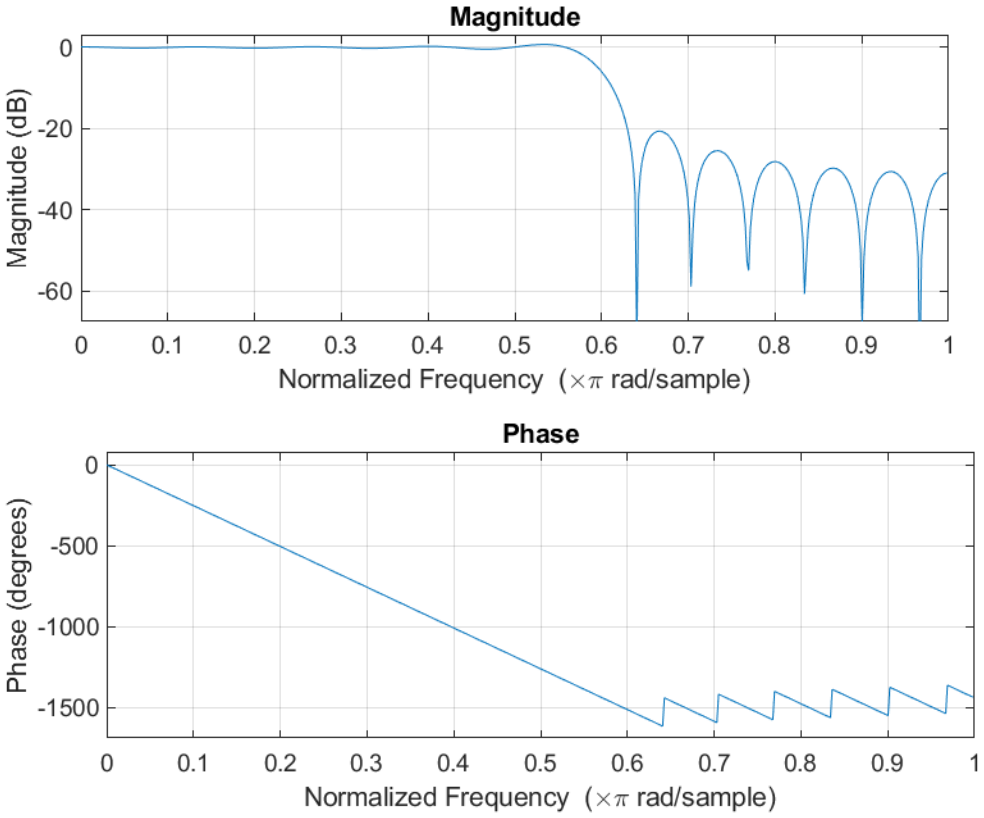

3. FIR filter design:

For low pass filter

Using Rectangle window:

```
%FIR filter design
N=input('enter the length of the filter')
wc=input('cutoff frequency=')
h=fir1(N,wc/pi,rectwin(N+1)) % response of filter
freqz(h)
```

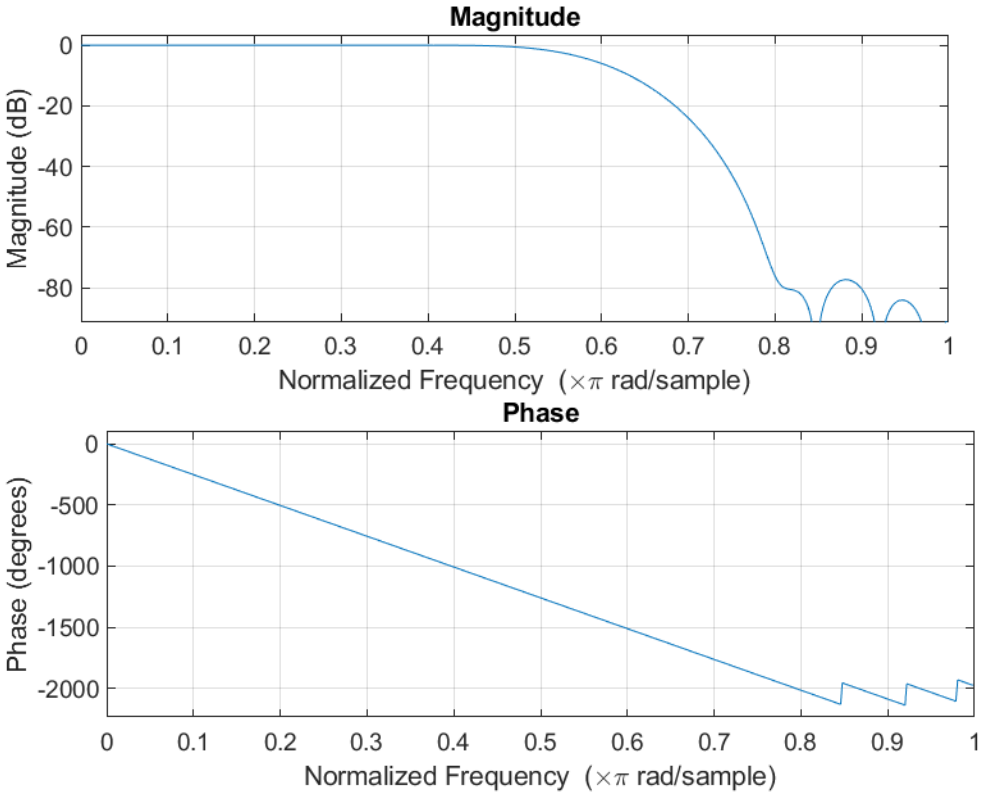
Input: N = 28; wc = 0.6π




Using Blackman window

```
%FIR filter design
N=input('enter the length of the filter')
wc=input('cutoff frequency=')
h=fir1(N,wc/pi,blackman(N+1)) % response of filter
freqz(h)
```

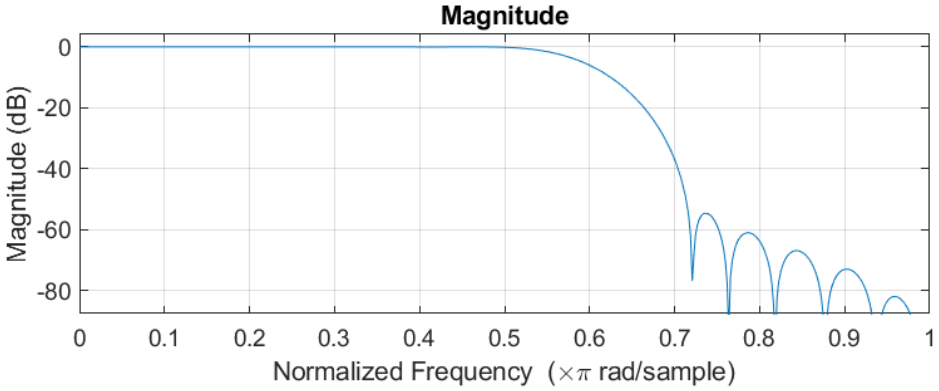
Input: N = 28; wc = 0.6π

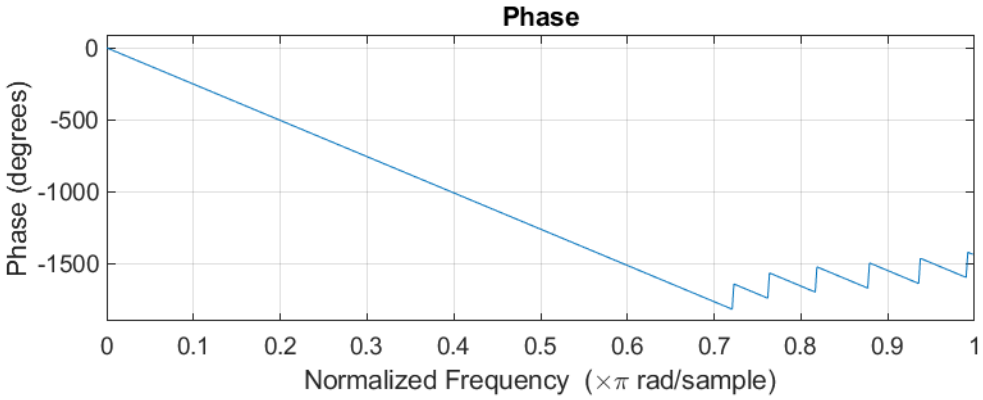



Using Kaiser window

```
% FIR filter design from kaiser window
N = input('Enter filter length: ');
wc = input('Cutoff frequency: ');
beta = 5; % for ~60 dB attenuation
h = fir1(N, wc/pi, kaiser(N+1, beta));
freqz(h)
```

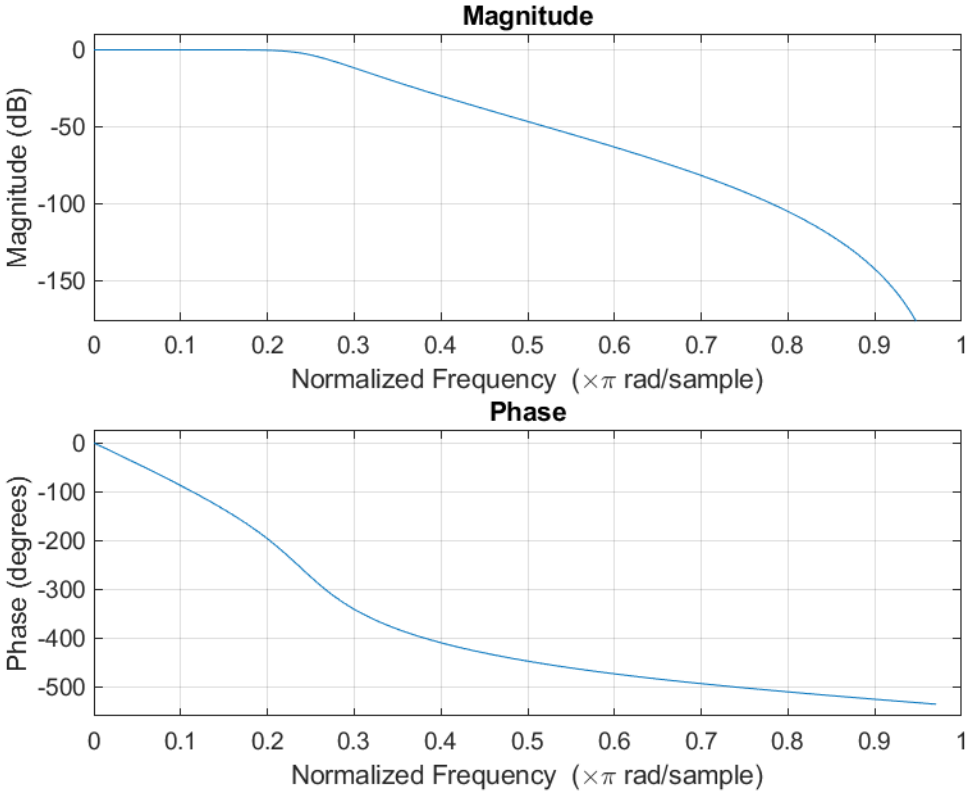
Input: N = 28; wc = 0.6$\pi$





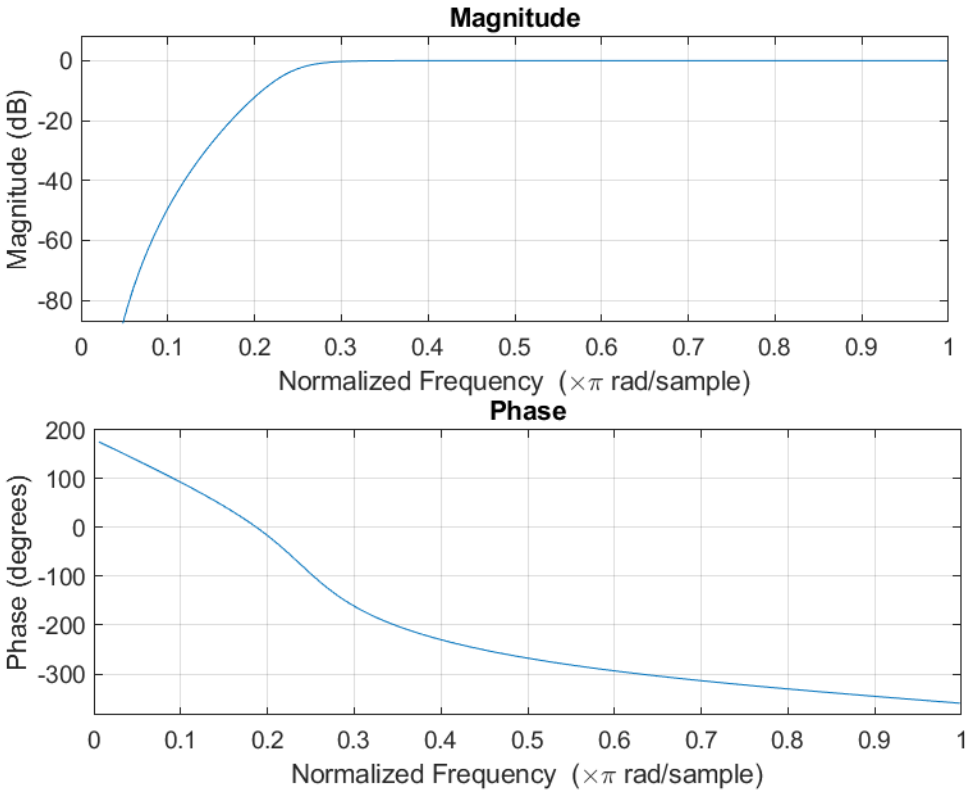

4. IIR filter design:

```
% IIR filter design
rp=0.4 % passband ripple
rs=30 %stopband attenuation
wp=0.2*pi %passband frequency
ws=0.4*pi %stopband frequency
[N,wn]=buttord(wp/pi , ws/pi, rp, rs) %finding order of the filter
[b,a]=butter(N,wn) %low pass
freqz(b,a)
```




Using same order for high pass filter

```
[c,d]=butter(N,wn, 'high') %high pass
freqz(c,d)
```




5. <u>FFT:</u>

For signal x[n] = `[0, 2, 1, 3]`

```
x2 = [0, 2, 1, 3];
N2 = length(x2);
X2 = fft(x2, N2);
subplot(4,1,1)
stem(0:N2-1, real(X2), 'filled', 'LineWidth', 1.5)
title('Real Part'); xlabel('k'); ylabel('Re{X(k)}'); grid on;
subplot(4,1,2)
stem(0:N2-1, imag(X2), 'filled', 'LineWidth', 1.5)
title('Imaginary Part'); xlabel('k'); ylabel('Im{X(k)}'); grid on;
subplot(4,1,3)
stem(0:N2-1, abs(X2), 'filled', 'LineWidth', 1.5)
title('Magnitude'); xlabel('k'); ylabel('|X(k)|'); grid on;
subplot(4,1,4)
stem(0:N2-1, angle(X2), 'filled', 'LineWidth', 1.5)
title('Phase'); xlabel('k'); ylabel('Angle (rad)'); grid on;
disp('Input x[n] = '); disp(x2)
fprintf('\nDFT X[k]:\n');
disp(X2.')
fprintf('\n k    Magnitude    Phase(rad)\n');
for k = 1:N2
    fprintf('%2d    %8.3f      %8.3f\n', k-1, abs(X2(k)), angle(X2(k)));
end
```

Output:

Input x[n] =

0 2 1 3

DFT X[k]:

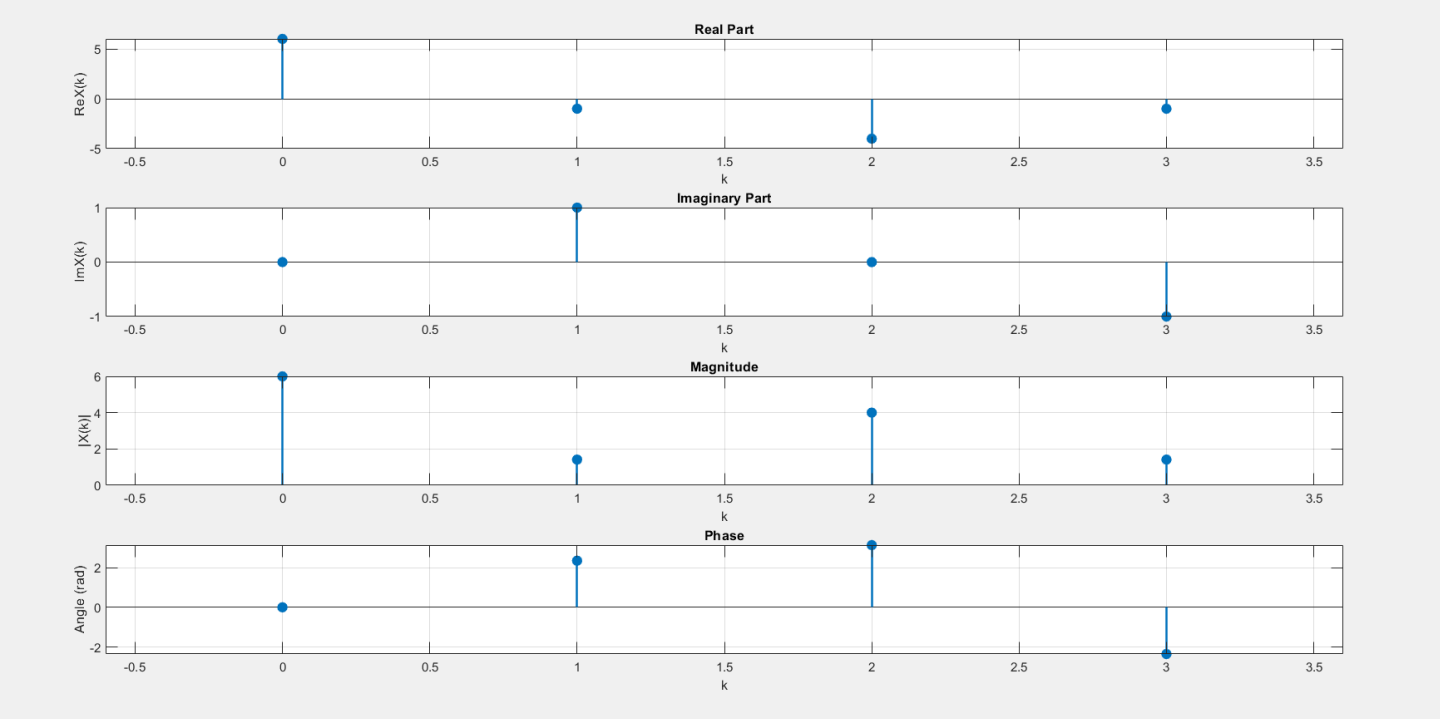
6.0000 + 0.0000i

-1.0000 + 1.0000i

-4.0000 + 0.0000i

-1.0000 - 1.0000i

| k | Magnitude | Phase(rad) |
|---|---|---|
| 0 | 6.000 | 0.000 |
| 1 | 1.414 | 2.356 |
| 2 | 4.000 | 3.142 |
| 3 | 1.414 | -2.356 |




For signal x[n] = `[1.2, 3.6, 2.4, -2, 1.6, 2.1]`

```
x1 = [1.2, 3.6, 2.4, -2, 1.6, 2.1, 0, 0];
```

```
N1 = length(x1);
X1 = fft(x1, N1);
subplot(4,1,1)
stem(0:N1-1, real(X1), 'filled', 'LineWidth', 1.5)
title('Real Part'); xlabel('k'); ylabel('Re{X(k)}'); grid on;
subplot(4,1,2)
stem(0:N1-1, imag(X1), 'filled', 'LineWidth', 1.5)
title('Imaginary Part'); xlabel('k'); ylabel('Im{X(k)}'); grid on;
subplot(4,1,3)
stem(0:N1-1, abs(X1), 'filled', 'LineWidth', 1.5)
title('Magnitude'); xlabel('k'); ylabel('|X(k)|'); grid on;
subplot(4,1,4)
stem(0:N1-1, angle(X1), 'filled', 'LineWidth', 1.5)
title('Phase'); xlabel('k'); ylabel('Angle (rad)'); grid on;
disp('Input x[n] = '); disp(x1)
fprintf('\nDFT X[k]:\n');
disp(X1.')
fprintf('\n k    Magnitude    Phase(rad)\n');
for k = 1:N1
    fprintf('%2d    %8.3f      %8.3f\n', k-1, abs(X1(k)), angle(X1(k)));
end
```

Output:

Input x[n] =

1.2000 3.6000 2.4000 -2.0000 1.6000 2.1000 0 0

DFT X[k]:

8.9000 + 0.0000i

2.0749 - 2.0464i

0.4000 - 7.7000i

-2.8749 + 2.7536i

1.5000 + 0.0000i

-2.8749 - 2.7536i

0.4000 + 7.7000i

2.0749 + 2.0464i

| k | Magnitude | Phase(rad) |
|---|---|---|
| 0 | 8.900 | 0.000 |
| 1 | 2.914 | -0.779 |
| 2 | 7.710 | -1.519 |

| | | |
|---|---|---|
| 3 | 3.981 | 2.378 |
| 4 | 1.500 | 0.000 |
| 5 | 3.981 | -2.378 |
| 6 | 7.710 | 1.519 |
| 7 | 2.914 | 0.779 |

